

Lec 10: Application layer: DNS

11/9/23

RECAP

• App layer protocol

• Transport layer used

HTTP	TCP
FTP	TCP
TFTP	UDP
SMTP	TCP
POP3, IMAP	TCP
DNS	UDP

* DNS (Domain Name System)

↳ client/server architecture

* UDP: transport layer
port # 53 ...

user defined

Client

port 53
server

* maps URL to IP address

key value store

Domain name	IP addr
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> trillions
of names

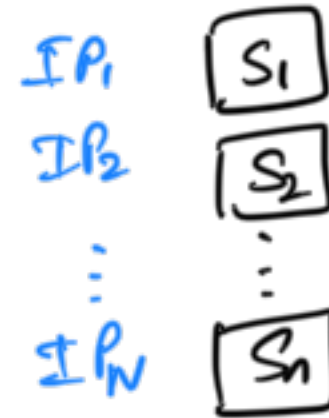
* centralized database X access time (large)

* distributed solution

- DNS services
 - * load balancing



www. amazon.in



Method 1

Client $\rightarrow$ Auth DNS server
 $\downarrow$
 returns any 1 addr out of N

Method 2

client $\rightarrow$ Auth DNS server
 $\downarrow$
 - returns all 'N' addr
 - client selects 1 addr

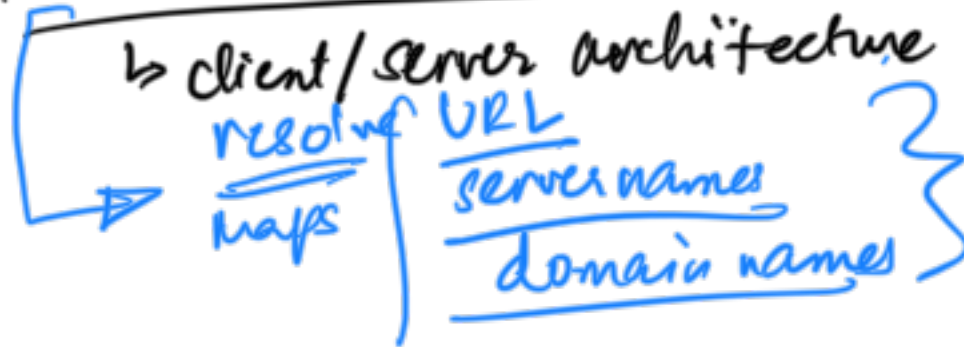
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⇒ IP address

• uses UDP protocol for transport

• port #: 63

* distributed solution
↳ hierarchical architecture for distributing data

